FOREST INTELLIGENCE

Drone Phenology Monitoring

End to end system of drone imagery collection, processing and phenology monitoring.

Description. Upload repeated drone orthomosaics of one site to detect, track and classify every tree crown and map its phenology.

About this service

Drone Phenology Monitoring is a web application that turns a time series of drone orthomosaics into a phenology map at individual tree-crown resolution. Each run is isolated: you create it, upload the orthomosaics, pick a model and parameters, and the pipeline runs end to end. Earlier runs stay available for comparison, so you can try different settings on the same site without losing anything.

Crown detection uses Detectree2 (Mask R-CNN). A graph-based tracker links the same crown across orthomosaic dates, and the phenology step uses those chains to decide whether a crown is deciduous and when its leaves fell and returned. When the run finishes, an interactive viewer opens the result as a clickable map: pick any crown and read its full phenology record on top of the orthomosaic.

How to use it

1. **Sign in and create a run.** Sign in with Google. From the dashboard, create a new run. Each run keeps its own uploads, parameters, logs and outputs, so you can compare runs later. The service does not re-run an existing run; parameter changes means a new run.

2. **Upload orthomosaics.** Add GeoTIFFs (`.tif` / `.tiff`) of the same site at different dates. Each file must carry a coordinate reference system in its metadata. The upload checks every file; if an orthomosaic is missing a CRS, you are prompted to select one before the pipeline starts.

3. **Pick a model and parameters.** Choose a Detectree2 weight (for example `250312_flexi` for mixed sites, or canopy specific weights). Set the tracking parameters, or keep the defaults.

4. **Run the pipeline.** Start the run. The four pipeline steps execute one after another, and the browser streams their logs live, so you can watch crown detection, tracking and phenology each finish in order. If you close the tab, the run keeps going on the server; come back later and the logs are still there with the finished outputs.

5. **Review and export.** Open the interactive viewer to inspect crowns and phenology overlays on top of the orthomosaic. Download the phenology vector (GeoPackage / GeoJSON), the per-crown CSV, the crown polygons, the pipeline log and the STAC Item from the per-run file browser.

Methodology

How Drone Phenology Monitoring works

> *Detectree2 (Mask R-CNN) · Graph-based tracker · Cloud-Optimised GeoTIFF · Leaflet.js · FastAPI · Airflow + STACD*

The pipeline runs four steps in sequence the moment you click *Upload & start pipeline*. Each step writes its outputs to a run-specific folder on the server, and the next step picks them up automatically.

Cards for the pipeline:

1 · Crown Detection

`01_crown_detection.py`

IN Orthomosaic GeoTIFFs OUT Per-date crown GeoPackages

A fine-tuned Detectree2 (Mask R-CNN) model outlines every visible tree crown in each orthomosaic. Overlapping detections are merged so each crown is counted once, with a confidence score per polygon.

2 · Crown Tracking

`02_crown_tracking.py`

IN Per-date crown GeoPackages OUT Consensus crowns GeoPackage

Crowns are linked across dates so each tree keeps one ID: anchor crowns align the dates by phase-correlation, then crowns are matched by IoU and centroid distance. Only those appearing in enough dates (`base_threshold`) survive, filtering out false detections and transient objects.

3 · Phenology Analysis

`03_phenology_analysis.py` · `10_phenophase_classifier.py` · `12_apply_phenophase_to_geojson.py`

IN Consensus crowns GeoPackage OUT Final annotated GeoJSON

A Green Chromatic Coordinate (GCC) time series is extracted per crown and reduced to a **Deciduousness Score (DS)** that classifies each crown as *deciduous* or *evergreen*. The label is written into the final annotated GeoJSON, the file the viewer loads and the primary deliverable for QGIS.

4 · Interactive Viewer

`04a_cog_tiling.py` · `04b_interactive_viz.py`

IN Orthomosaics + Final GeoJSON OUT Self-contained Leaflet map HTML

Orthomosaics are tiled into Cloud-Optimised GeoTIFFs so the viewer loads only the tiles it needs, then assembled with the annotated GeoJSON into a self-contained Leaflet.js map: clickable crowns, a date slider across acquisition dates, and a switch between the orthomosaic and a false-colour vegetation view.

Models and parameters

Detector (Detectree2 / Mask R-CNN). Instance segmentation with published weights, not retrained per site. Three weight options are selectable at run time:

MODEL VARIANT	WHEN TO USE
`urban_canopy` *(default)*	Mixed urban / peri-urban canopy; well-separated crowns
`250312_flexi`	General-purpose; a good first fallback if urban_canopy underfits
`tropical_dense`	Dense tropical closed-canopy forests with heavy crown overlap

Tracking parameters. `base_threshold` (default 45) sets how many dates a crown must appear in to be kept. Raise it to drop more transient detections, lower it to retain more. `align_threshold` (default 65) sets the confidence floor for anchor crowns used in alignment. `align_method` defaults to `pcc_tiled` (phase-correlation on tiles).

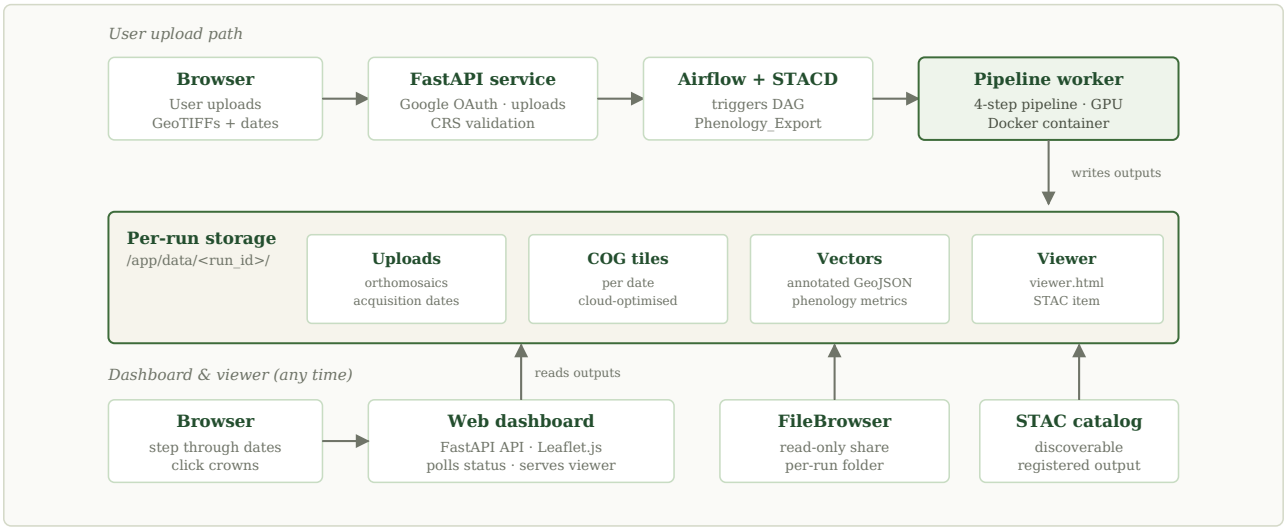
Deciduousness Score weights. Four weights (`w_veg_amp` , `w_depth` , `w_gcc_amp` , `w_tex`) combine the four phenological signals into a single DS used by the classifier. Defaults are tuned for mixed temperate / subtropical forests; the Parameter Guide covers tuning for other biomes.

System architecture

- **Pipeline worker computes, FileBrowser stores, dashboard only reads.** The three layers are decoupled: a run's results, once written, do not depend on the pipeline worker to be served.
- **Upload path:** drone orthomosaics (GeoTIFFs with embedded CRS) are uploaded via the browser to the FastAPI service, which validates the CRS, writes the files, and records acquisition dates against them in the database.
- **Pipeline worker:** runs the four-step pipeline inside Docker. Writes outputs to `/app/data/<run_id>/outputs` . GPU inference is used for the detection model.
- **Scheduling:** on upload complete, the FastAPI service POSTs a trigger to Airflow for the `drone_phenology_monitoring` DAG with the `run_id` plus any overridden parameters. The DAG runs the `Phenology_Export` root algorithm, which calls the pipeline's `/api/export-phenology` endpoint and reads its JSON reply.
- **Storage:** per-run folders under `/app/data/<run_id>/` hold uploads, intermediate outputs and the final viewer. FileBrowser provides read-only share links for each run folder.
- **Web dashboard:** a FastAPI JSON API serves run metadata, triggers runs, and streams per-step status. A browser front end built on Leaflet.js renders the per-run interactive viewer, the dashboard polls run status every five seconds, and signed-in users see only their own runs.
- **Standalone viewer:** each run also produces a self-contained `viewer.html` that references its tiles with relative paths. It

can be opened in any browser without the DPM backend.

The user uploads once; the worker runs the pipeline; the dashboard and viewer only read



The top row is a one-time run per site. The bottom row is what happens whenever someone opens the dashboard or the viewer. Per-run storage sits between them.

Outputs

- **Interactive viewer (`viewer.html`):** self-contained Leaflet web map with a date slider, clickable crown polygons, tooltips showing tracked ID and phenophase, and a base-map switch between the orthomosaic and a false-colour vegetation view.
- **Final annotated GeoJSON (`final_annotated_crowns.geojson`):** all tracked crowns with the per-crown phenophase label and crown geometry, the primary deliverable for QGIS.
- **Phenology metrics GeoJSON (`phenology_metrics.geojson`):** raw metrics before classification, useful for independent analysis of the fitted curves.
- **COG tiles (`cogs/`):** one Cloud-Optimised GeoTIFF per acquisition date, reusable as base layers in any COG-aware GIS tool.
- **STAC Item:** the final per-crown vector is registered automatically by STACD (dataset type `DPM_Phenology_Vector`, table extension) so the output is discoverable in the shared catalog.
- **Public landing-page listing:** toggle a run *Public* and its viewer appears on the DPM landing page for anyone to open without signing in.
- **Standalone viewer:** the viewer HTML and its tiles are self-referencing with relative paths, so the whole run folder can be moved and the viewer still opens.

Limitations

- **Detection depends on canopy visibility.** Understorey trees obscured by taller crowns are not detected. The pipeline operates on nadir drone imagery only; oblique captures are not supported.
- **Phenology metrics need temporal coverage.** At least five dates spread across the active season are required for reliable curve fits; eight to twelve dates spaced two to three weeks apart are ideal. Missing either end of the season (leaf-out or leaf-off) degrades phenophase accuracy.
- **Alignment assumes spatial overlap.** Tracking links crowns across dates using phase-correlation alignment on anchor crowns. A survey that flew a very different path on one date may fail alignment for that date.
- **Classification is sensitive to DS weights.** Defaults are tuned for mixed temperate / subtropical forests; sites outside that envelope may need the Parameter Guide's tuning procedure.
- **Detection weights are not retrained.** Published Detectree2 weights are used as-is, so detection quality depends on how close your site is to the training distribution. Picking the closest variant for your canopy type matters.
- **Illumination consistency matters.** Shadows from low sun angles introduce spectral noise that degrades the phenological signal. Fly within ±2 hours of solar noon on every acquisition date.
- **Boundary consistency matters.** Expanding the survey polygon across dates changes which crowns are detected and tracked. Reuse the exact saved flight plan every date.